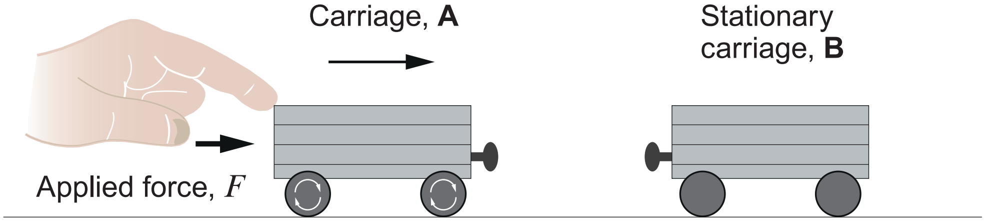
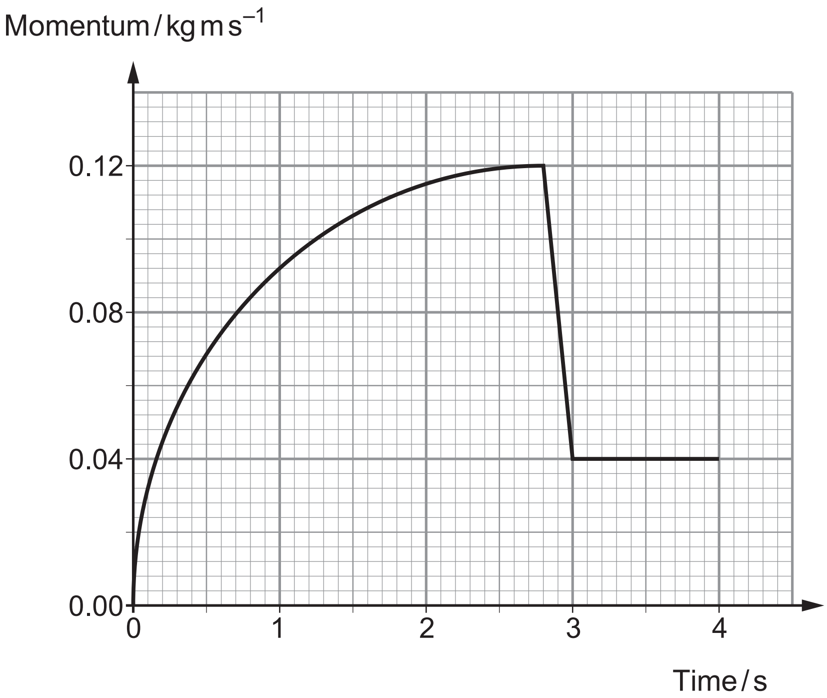
Examiner only

6. A toy train carriage, **A**, is accelerated from rest by an applied force in a straight line on a smooth track towards a stationary carriage, **B**. A graph of momentum against time for carriage **A** is shown below.





(a) Estimate, from the graph, the resultant accelerating **force** acting on carriage A at a time of 1.0 s. [3]

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(b) Impact between carriage A and carriage B occurs between 2.8 s and 3.0 s. Just before impact the applied force is removed. State how the graph confirms that the applied force is removed, and explain why its removal enables you to calculate the momentum given to carriage B. [3]

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(c) (i) **Draw, on the graph opposite**, a line showing the momentum of **carriage B** between 0 s and 4 s. [3]

(ii) Hence calculate the speed of **carriage B** after impact given that its mass is 0.16 kg. [2]

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